Project Report

Only for course Teacher						
		Needs Improvement	**Developing**	**Sufficient**	**Above Average**	**Total Mark**
Allocate mark & Percentage		**25%**	**50%**	**75%**	**100%**	**25**
Problem understanding & Analysis	**7**					
Implementation	**8**					
Report writing	**10**					
Total obtained mark						
Comments						

Semester: Spring 2026

Batch: 40 **Section:** E2

Course Name: Software Engineering Design Capstone Project

Course Code: SE-331

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Designation: Lecturer, Department of Software Engineering, DIU

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Submission Date: 08/02/2026

Project Proposal: BD-Vote

Subtitle: A Secure and Transparent E-Voting Support System for Bangladesh using NID-Based Digital Verification

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1. Introduction

This project proposes **BD-Vote**, a digital platform designed to make the election process in Bangladesh more modern, transparent, and secure. It focuses on reducing human intervention in voter verification and result counting by relying on advanced technology.

1.1 Background Overview

Application Domain: Smart Governance & Secure E-Voting Systems

Conducting elections in a democratic country like Bangladesh is a massive and expensive task. In the 2024 national election, a huge portion of the budget (over 2,000 Crore BDT) was spent on law enforcement and manpower management. Currently, paper ballots or EVMs require voters to be physically present, which is difficult for expatriates or sick citizens. Moreover, there are often transparency concerns. BD-Vote aims to solve these issues using AI and Distributed Ledger (Blockchain) technology.

1.2 Problem Statement

The current election system faces several major issues:

- **High Cost and Manpower:** Thousands of security personnel and officers are needed at every center, requiring a massive budget.
- **Booth Jamming & Proxy Voting:** Issues like "booth capturing" or people casting someone else’s vote obstruct the democratic process.
- **Lack of Transparency:** Since data is stored in central databases, there is a risk of unauthorized changes during result counting.
- **Voter Hesitation:** Many conscientious citizens avoid voting centers due to long queues and fear of chaos.

BD-Vote allows voters to vote securely from home using NID and Face Verification, completely eliminating the risk of booth capturing.

1.3 Objectives

- **General Objective:** To develop a decentralized, AI-powered e-voting platform for Bangladesh that ensures 100% transparency and accurate voter identification.
- **Specific Objectives:**
- Identify fake voters using NID and Face-API.
- Prevent double voting (one person, one vote).

- Store votes using Blockchain to make the data unchangeable (tamper-proof).
- Provide real-time election results on a public dashboard.

1.4 Scope

- In-Scope:
    - NID verification via OCR
    - live face detection and matching
    - secure digital ballot
    - and a real-time results dashboard
- Out-of-Scope:
    - Managing physical law and order at polling stations
    - creating new NID registrations
    - and voting for individuals under 18

1.5 Stakeholders

BD-Vote involves several key stakeholders who ensure the election process is functional, secure, and transparent:

- **Voters:** The primary users of the system. These are citizens (18+ years old with valid NIDs) who will perform NID verification and live face matching to cast their votes securely from any location.
- **Election Commission (EC):** The central administrators of the platform. They are responsible for managing the candidate list, uploading election symbols, setting voting timelines, and officially announcing the final results.
- **Law Enforcement Agencies:** They act as the "Digital Guardians" of the system. Their role is to monitor backend security, prevent cyber-attacks (like DDoS or hacking), and investigate any flagged fraudulent activities or multiple login attempts from suspicious sources.
- **Independent Observers:** These include local and international bodies, as well as the media. They use a public blockchain dashboard to audit the transparency of the election and verify that the results are tamper-proof without accessing private voter data.
- **Technical Team (Developers):** The engineers responsible for maintaining the system’s uptime, managing the Supabase database, and ensuring the Blockchain ledger and AI modules (OCR and Face-API) are functioning correctly during the election.
- **Candidates:** The direct participants of the election. Each candidate has access to a dedicated dashboard to monitor real-time voter turnout and the specific vote count in their respective constituencies.

1.6 Proposed Solution

BD-Vote follows a multi-layer security protocol. First, the voter scans their NID, which is verified via OCR. Then, the system performs face-matching. Once verified, the voter casts their vote for their preferred candidate, and that data is stored in an encrypted ledger. This removes manual counting errors and the possibility of fraud.

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2. System Requirements

2.1 Functional Requirements (FR)

- **FR1 (Voter Authentication):** The system must allow voters to be identified via NID number and OCR scanning.
- **FR2 (Identity Validation):** The system must match the voter's live face with their NID photo using Face-api.js.
- **FR3 (One-Vote Constraint):** The system must ensure a voter cannot cast more than one vote.
- **FR4 (Secure Vote Casting):** After selection, the vote must be stored in the database in an encrypted format.
- **FR5 (Real-time Aggregation):** The system must count votes automatically and show real-time results on a public dashboard.
- **FR6 (Admin Management):** Admins must be able to manage candidates, symbols, and constituency data.

2.2 Non-Functional Requirements (NFR)

- **Security:** Voting data must be stored on a Distributed Ledger to make it **Immutable** (unchangeable).
- **Scalability:** The system must handle high traffic from thousands of voters simultaneously.
- **Transparency:** Every vote must be auditable while keeping the voter's identity private.

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3. Tools and Technologies

- **Frontend:** Typescript and Tailwind CSS for a responsive UI.
- **AI & Computer Vision:** Face-api.js for facial recognition and Tesseract.js for NID OCR.
- **Blockchain & Security:** Solidity for Smart Contracts and Ethers.js for blockchain interaction.
- **Backend & Database:** Supabase (PostgreSQL) for metadata and a **Distributed Ledger** for vote records.

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4. Project Timeline and Work Plan

Milestone	Task Description	Duration	Responsibility
M1	**Requirement Analysis:** Finalizing system architecture, UI wireframes, and database schema.	**Week 1**	Full Team
M2	**Backend & Core Setup:** Setting up Supabase, NID metadata tables, and Auth API.	**Week 2**	Shamimul Islam Sunny
M3	**AI Module Development:** Implementing Tesseract.js (OCR) and Face-api.js (Matching).	**Week 3–4**	Rafia Tasnim Risha
M4	**Blockchain & Smart Contracts:** Developing Solidity contracts and testing voting logic on Hardhat.	**Week 3–5**	Fahim Bin Forhad
M5	**System Integration:** Connecting the AI module and Blockchain layer with the Next.js Frontend.	**Week 6**	Full Team
M6	**Frontend Refinement:** Finalizing the Admin Dashboard and Public Results Analytics.	**Week 7**	Full Team